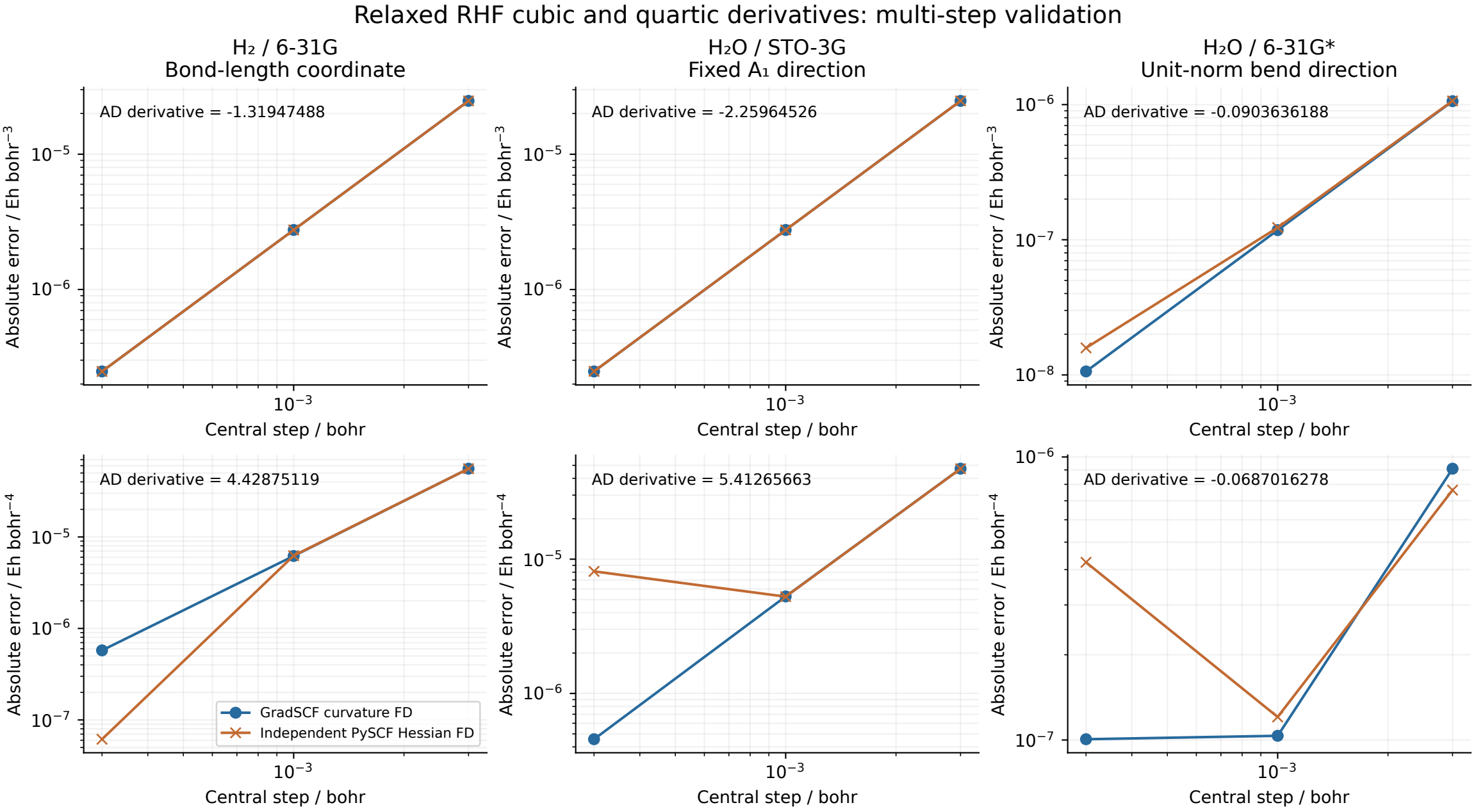
Relaxed RHF cubic and quartic derivatives: multi-step validation
H₂ / 6-31G
Bond-length coordinate
H₂O / STO-3G
Fixed A₁ direction
H₂O / 6-31G*
Unit-norm bend direction
AD derivative = -1.31947488
AD derivative = -2.25964526
AD derivative = -0.0903636188
AD derivative = 4.42875119
AD derivative = 5.41265663
AD derivative = -0.0687016278
Absolute error / Eh bohr⁻³
Absolute error / Eh bohr⁻⁴
Central step / bohr
GradSCF curvature FD
Independent PySCF Hessian FD